

EXISTING SOLUTIONS & RESEARCH AND SELECTED PS

Search Is Evolving—Tools Haven't



Phase 1 – Traditional SEO (Macro Layer)

- **Tools:** Ahrefs, SEMrush
- ✗ **Optimizes:** Links & Rankings



Phase 2 – Content Ops (Micro Layer)

- **Tools:** SurferSEO, Clearscope
- ✗ Encourages keyword stuffing
- ✗ **Reduces AI trust**



Phase 3 – GEO (IMPERION) (Synthesis Layer)

- **Optimizes:** Trust & Citability
- ✓ How AI selects & rewrites content
- ✓ Focuses on being the **source**, not ranking

Links → Keywords → Trust & AI Synthesis



Research Gap

- No tool analyzes AI-generated answers
- No visibility into why AI includes or ignores content



Selected Problem Statement (PS01)

PS01: Building a Generative Engine Optimization (GEO)
Platform for AI-Driven Search & Discovery

Modern AI engines generate **one answer**, not ten links – yet websites have zero transparency into this process.



Traditional SEO
is link-based



New AI search
gives one answer



New AI search
gives one answer



No tools exist
to analyze AI answers



IMPERION

SOLUTION APPROACH

Step 1: Digital Twin Input

System accepts Target URL + User Intent (Query).

Purpose: Establishes the "Source" and the "Question".

Step 2: Structural Analysis (The Spy)

Scrapes content to extract the "Skeleton": H-Tags, Word Count, and Schema.

Differentiation: We ignore keywords and focus on Structural Signals.

Step 3: AI Simulation Engine

Queries Google Gemini to generate the "Ground Truth" answer.

Value: Simulates exactly how a modern engine synthesizes the topic in real-time.

Step 4: Synthesis View

Read-only display of the AI's answer with Format Detection (List vs. Paragraph).

Purpose: Radical transparency into the AI's "Mental Model".

Step 5: GEO Analysis Layer (Core)

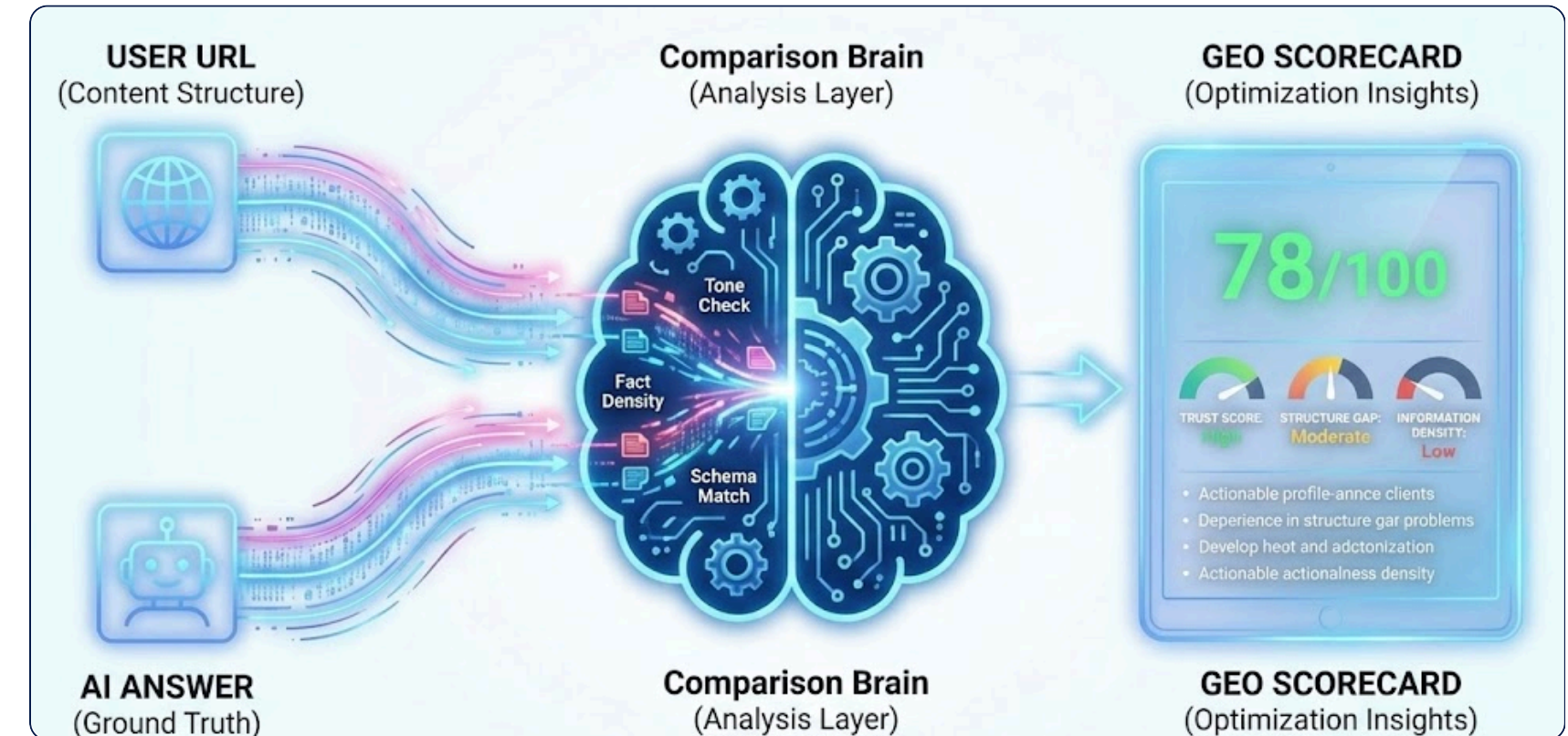
We evaluate content using 4 smart checks:

- Hype-Checker: Penalizes salesy language (0–100 score)
- Fact Density: Facts vs fluff ratio
- Readability Match: Aligns with AI reading level
- Schema Check: Lists & tables preferred by AI

Step 6: Optimization Engine

Generates specific fixes: "Add a Comparison Table,"

"Reduce Promotional Tone," "Increase Fact Density by 4%".



UNIQUE FEATURES AND INNOVATION

Features	Traditional SEO / Content Ops (Ahrefs, SurferSEO)	IMPERION (GEO - Our Solution)
Core Approach	Relies on Keyword Metrics & Backlinks	AI Answer Simulation
Analysis Logic	Keyword Density (Counting words)	Structure-Based Comparison (Headings, Depth, Format)
The Output	Raw Analytics (Charts & Data tables)	Actionable Recommendations (Specific fixes for synthesis)
Tone Check	Encourages "Salesy" Copywriting (To convert humans)	Trust Score Algorithm (Penalizes promotional hype)
Quality Metric	Word Count (More words = Better)	Fact-Density Analysis (More Entities = Higher Quality)

THE "TRUST SCORE" (HYPE-CHECK)

The Problem:

Traditional tools encourage promotional language ("Best," "Cheap," "Buy Now"). LLMs like Gemini operate on E-E-A-T principles (Experience, Expertise, Authoritativeness, Trust) and discard this as "marketing fluff".

Our Solution:

We calculate a 0-100 Trust Score. We penalize "Salesy" language and reward objective, encyclopedia-style neutrality to increase citation probability

DYNAMIC "FACT-DENSITY" ANALYSIS

The Problem:

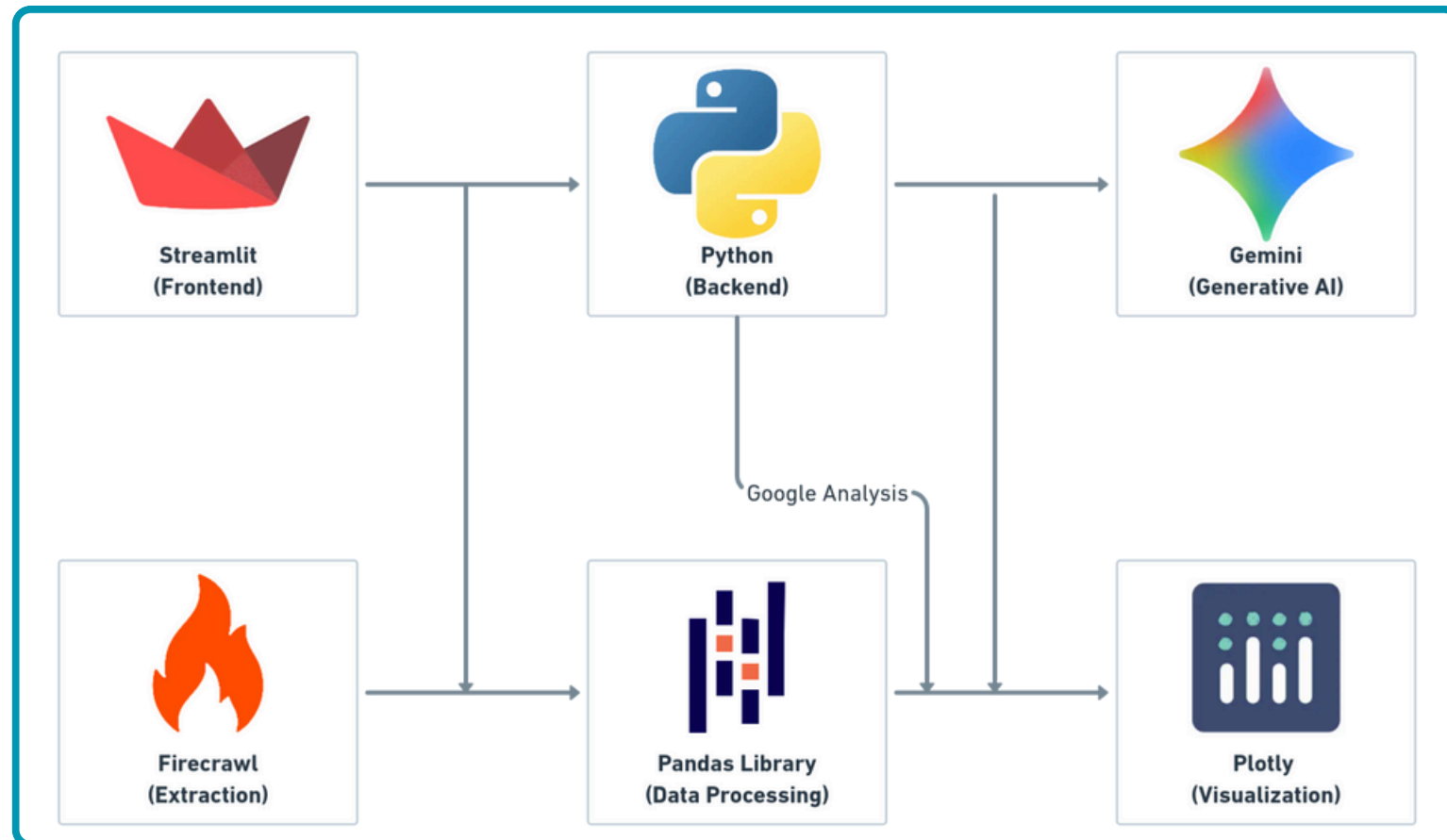
Competitors give vague advice like "Write better content." They measure quality by length (Word Count), leading to fluff.

Our Solution:

Imperion is the first to quantify Information Density. We calculate the ratio of Hard Entities (Names, Dates, Statistics) to total words. This gives users a specific mathematical target (e.g., "Increase Entity Density by 4%").

TECHNOLOGY STACK & POTENTIAL IMPACT

Tech Stack



Who Benefits

- Startups - Gain visibility in AI discovery platforms
- Bloggers/Creators - Earn AI citations vs. fighting for backlinks
- Businesses - Optimize knowledge bases & documentation

Key Outcomes

- Bridges the gap between AI-generated answers & website content
- Saves hours of manual AI-behavior guesswork
- Prepares content for the zero-click, synthesis-based search era

Key Outcomes

